

Skill Check 2: Exploring Data

Exam Agreement

This Skill Check is an individual assessment and you should not receive or offer help on it from any other human. Cell phones are to be handed in to the instructor and all other digital devices must be stored in bags underneath your desk or against the wall.

You may use any **physical** resource to complete the work. This includes:

- Any notes, code, slides, papers, or previous feedback from the instructor as long as they are on paper.
- Any books that you have with you.
- Any scholarly works such as papers that you have with you.

You may NOT use:

- Help from any other student or person. **This is an individual assessment.**
- Any digital resource that does not exist as a physical copy present in class.
- The use of generative artificial intelligence (e.g., ChatGPT).
- Help from homework websites such as Course Hero or Chegg.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Section: _____

Signed: _____

Date: _____

Include this signed page as the first page of your submitted work.

Skill Check Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: Loading, Cleaning, and Analyzing Data

You have a data file that you'd like to analyze in R named `tiger_tails.csv` that contains measurements of tiger swallowtail butterfly wings. It consists of an ID number (`number`), measurements of wing length (`wing_length`), width (`wing_width`), and the field site (`site`). The first line of this file contain information on the dates and times butterflies were captured and measured but do not contain the measurements that you wish to analyze.

- a. Write the code to load the data set into R as a data frame named `tiger_tails`.

- b. What types of data are each of these columns (except ID number)? (Note: asking for general data type relevant to analysis, not the data class.)

- c. If you wanted to plot wing length versus field site, what would be the best type of plot for visualizing these data?

- d. Calculate the mean value of wing width in the data set for each field site: **Boone, Asheville, and Knoxville.**

Question 2: Plotting Basics

Using the Blackmore data set, make a scatter plot of exercise versus age and color the points by group.

 Blackmore	945 obs. of 4 variables	
\$ subject : Factor w/ 231 levels "100","101","102",...: 1 1 ...		
\$ age : num 8 10 12 14 15.9 ...		
\$ exercise: num 2.71 1.94 2.36 1.54 8.63 0.14 0.14 0 0 5.0...		
\$ group : Factor w/ 2 levels "control","patient": 2 2 2 2...		

a. With base graphics.

b. With ggplot2 package.

Question 3: Visualizing and Communicating Data

Describe the ethical obligation you have when visualizing data. Why is this important?